3.2 Law of Cosines

Name _____ Period ____ Date _____

Pre-Calculus 3.2 (30-minute plan) · Student sheet

I can…

- use the Law of Cosines to solve oblique triangles (SSS or SAS).
- decide whether to use the Law of Sines or the Law of Cosines.
- use Heron's Area Formula and solve real-life problems.

Vocabulary · fill in the missing word

Law of Cosines — $a^2 = b^2 + c^2 - 2bc \cos A$ (and the same _____ for b^2 and c^2)

Heron's Area Formula — Area = _____, where $s = \frac{a + b + c}{2}$

Semiperimeter — s = _____

Part 1 · Notes and examples (fill in each note, then work the example)

Law of Cosines: c^2 = _____

Solving for an angle: To find an angle: cos C = _____

SSS: For SSS, find the _____ angle first.

Example 1

Solve the triangle: a = 6, b = 8, c = 12. Round to two decimal places.

Example 2

Solve the triangle: A = 115°, b = 15, c = 10. Round to two decimal places.

Which law first?: Use the Law of Cosines first for _____ and _____.

Heron's Area Formula: s = _____, Area = _____

Example 3

Find the area of a triangle with sides a = 43 m, b = 53 m, and c = 72 m.

Part 2 · Partner practice

Partner practice

C = 40°, a = 9, b = 12. Find c. Round to two decimal places.